IGCSE 0607 Practice Questions — E3.6 — Combined Questions & Markscheme

Transforming graphs of functions

Total Questions: 3 — **Total Marks:** 18 Marks

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  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(1 Q)*

1 A function is defined as $f(x) = 2x - 5$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find $g(x)$ in terms of x, giving your answer in its simplest form.

$g(x) =$ ..... **[2]**

(b) Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = f(x - 4)$.

.....

.....

..... **[2]**

2 Let $f(x) = 3x^2$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ to give the graph of $y = g(x)$.

Find $g(x)$ in terms of x.

Give your answer in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

(b) The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 0 \\ -8 \end{pmatrix}$ to give the graph of $y = h(x)$.

Write down the coordinates of the minimum point of the graph of $y = h(x)$.

(..... , ...) **[2]**

(c) Find the y-intercept of the graph of $y = h(x)$.

..... **[1]**

3 A function is defined as $f(x) = \dfrac{20}{x}$.

(a) The graph of $y = f(x)$ is mapped onto the graph of $y = g(x)$ by a translation with vector $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$.

(i) Write down an expression for $g(x)$.

$g(x) =$ ..... **[1]**

(ii) Solve the equation $g(x) = 0$.

$x =$ ..... **[2]**

(b) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$ to give the graph of $y = h(x)$.
The graph of $y = h(x)$ passes through the point $(8, 5)$.
Work out the value of k.

$k =$ ..... **[3]**

(c) Let $p(x) = f(x - 3) + 7$.
Write down the equations of the vertical and horizontal asymptotes of the graph of $y = p(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

1 **(a)** $g(x) = f(x) + 7$ **[M1]**

$g(x) = 2x + 2$ **[A1]**

(b) Translation **[B1]**

by vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ **[B1]**

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2 **(a)** $g(x) = f(x + 2) = 3(x + 2)^2$ **[M1]**

$3(x^2 + 4x + 4)$ **[M1]**

$3x^2 + 12x + 12$ **[A1]**

(b) $(-2, -8)$ **[B2]**

(Allow B1 for each correct coordinate)

(c) Substitutes $x = 0$ into $h(x) = 3(x + 2)^2 - 8$ (or similar valid method) **[M1]**

4 **[A1]**

[Back to Question] *[Total Marks: 6]*

3 **(a)** **(i)** $\frac{20}{x} - 4$ **[B1]**

(ii) $\frac{20}{x} - 4 = 0$ **[M1]**

$x = 5$ **[A1]**

(b) $h(x) = f(x - k) = \frac{20}{x - k}$ **[M1]**

$\frac{20}{8 - k} = 5$ **[M1]**

$k = 4$ **[A1]**

(c) Vertical asymptote: $x = 3$ **[B1]**

Horizontal asymptote: $y = 7$ **[B1]**

[Back to Question] *[Total Marks: 8]*